Prompt Engineering for Agents

50+ Battle-Tested Prompts

ErgeneAI

Ch 1: System Prompts for AI Agents

12 battle-tested system prompts for different agent roles.

1. General Assistant

You are a helpful AI assistant. Be concise, accurate, and proactive.

Always verify infor

2. Customer Support Agent

You handle customer inquiries. Be empathetic and solution-oriented.

Stay within your kn

3. Data Extraction Agent

Extract structured data from unstructured text.

Output only valid J

4. Code Generator

Generate production-ready code. Add comments for complex logic.

Prefer readability o

5. Content Writer

Write in active voice. Short sentences. One idea per paragraph.

Match the tone req

6. Decision Engine

Analyze options based on: feasibility, cost, time, risk.

Provide weighted r

7. Quality Checker

Review content for: accuracy, clarity, completeness, consistency.

List issues by seve

8. Email Composer

Keep emails under 150 words. Clear subject line. One call to action.

Professional but no

9. Social Media Manager

Platform-aware content adaptation. Use relevant hashtags.

Engagement hooks

10. Research Assistant

Cite sources. Distinguish fact from opinion.

Summarize key fin

11. Workflow Designer

Design n8n workflows. Define triggers, actions, error handling.

Output as JSON w

12. Debug Agent

Identify root cause. Propose minimal fix. Verify before confirming.

Ch 2: Chain-of-Thought Templates

10 templates for structured reasoning.

1. Problem Decomposition

Break the problem into: inputs, constraints, desired outputs.

Solve each sub-pro

2. Root Cause Analysis

Symptom > Possible causes > Evidence check > Root cause > Fix

List 3+ possible ca

3. Decision Matrix

Options > Criteria > Weights > Scores > Recommendation

Score each option

4. Cost-Benefit

List costs (time, money, risk). List benefits (direct, indirect).

Quantify where pos

5. Step-by-Step Plan

Goal > Prerequisites > Steps > Verification > Rollback

Each step must be

6. Pro-Con Analysis

For each option: list 3 pros, 3 cons, 1 risk.

Weight by probabil

7. What-If Scenarios

Best case, worst case, most likely case.

For each: trigger, c

8. Learning from Failure

What happened. Why it happened. What was learned.

How to prevent rec

9. Trade-off Analysis

Two competing priorities. Quantify each.

Find optimal balan

10. Reverse Engineering

Observed output > Deduce process > Identify components.

Recreate logic. Ve

Ch 3: Tool-Use Instructions

10 patterns for effective tool use.

1. Single Tool Execution

Use the tool. Get result. Decide next step.

Do not plan tool ca

2. Multi-Tool Sequence

Tool A output > Tool B > Combine results.

Each call depends

3. Search-Then-Act

Search first. Understand context. Then act.

Never act without s

4. Read-Then-Write

Read existing content. Understand structure.

Then write. Match

5. Verify-Before-Confirm

Before confirming: verify output.

Check: format, con

6. Error Recovery

On error: log it. Understand it. Retry or abort.

Never silently igno

7. Rate Limit Handling

On rate limit: wait. Retry with backoff.

Log how many retr

8. Pagination Pattern

Request 1 page. Process. Check for next.

Continue until all p

9. Conditional Logic

If true: path A. If false: path B.

Always handle both

10. Batch Processing

Split into chunks. Process each. Combine.

Track progress: X

Ch 4: Error Recovery Prompts

8 patterns for graceful failure handling.

1. Retry Prompt

The previous attempt failed. Analyze the error.

If transient: retry w

2. Degradation Prompt

Primary approach failed. Use fallback method.

Log: what failed, w

3. User Clarification

I could not complete this task because: [reason].

To proceed I need

4. Partial Results

Completed X of Y tasks. Partial results available.

Failed tasks: [list].

5. Timeout Recovery

Operation timed out. Save partial state.

Resume from chec

6. Context Recovery

Context was lost. Key facts: [summary].

Last completed ste

7. Alternative Strategy

Primary strategy failed due to: [reason].

Switching to: [alter

8. Graceful Shutdown

Unrecoverable error. Saving state.

Report: what was

Ch 5: Multi-Agent Coordination

10 patterns for coordinating multiple AI agents.

1. Leader-Follower

Leader divides task. Followers execute. Leader integrates.

Each follower has

2. Publisher-Subscriber

One agent publishes output to channel.

Other agents subs

3. Review Loop

Agent A produces. Agent B reviews. Agent A revises.

Max 3 revision cyc

4. Voting

Multiple agents solve same problem independently.

Compare results. U

5. Pipeline

Sequential processing. Each agent transforms output.

Well-defined interfa

6. Fan-Out

One coordinator. Multiple parallel workers.

Workers independe

7. Escalation

Agent A handles routine. Escalates edge cases to Agent B.

Clear criteria for wh

8. Consensus Building

Agents debate. Resolve disagreements.

Document: disagre

9. Specialized Roles

Each agent has single expertise. Router delegates.

Router: intent class

10. Checkpoint-Review

Agent completes checkpoints. Reviewer approves.

Reviewer can: app